SHUBHAM GAJJAR

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EDUCATION

Northeastern University, Portland, Maine — September 2025 – May 2027
Master of Science in Artificial Intelligence, Grade Point Average: 4.0/4.0

LDRP Institute of Technology and Research, Gandhinagar, India — September 2022 – May 2025
Bachelor of Engineering in Computer Engineering, Grade Point Average: 8.41/10.0

Relevant Coursework: Foundations of Artificial Intelligence, Machine Learning, Research Capstone, Applied Programming and Data Processing for AI, Deep Learning, Computer Vision, Algorithms, Image Processing

TECHNICAL SKILLS & CERTIFICATION

Programming Languages: Python, JavaScript
Deep Learning: PyTorch, TensorFlow, Keras, HuggingFace Transformers, CUDA, QLoRA
Computer Vision & Multimodal: OpenCV, Albumentations, Matplotlib, MedGemma, MedSigLIP, DINOv3, CLIP
Data Science & MLOps: NumPy, Pandas, Scikit-learn, Seaborn, Jupyter, Databricks, MLflow, Unity Catalog, Spark, Docker, PostgreSQL
Web Development & Tools: Next.js, React Native, FastAPI, Tailwind CSS, Flask, Git, Vercel
Certificates: Python for Data Science from Indian Institute of Technology Madras, Python Data Structures from University of Michigan

PROFESSIONAL EXPERIENCE

Northeastern University, Portland, Maine
Research Assistant — February 2026 – Present
- Designed end-to-end VLM fine-tuning pipeline using MedGemma 1.5 4B with QLoRA on MedSigLIP encoder for veterinary mast cell tumor classification across 9 grade categories from fine needle aspirate cytology
- Curated 5-task VQA dataset from hierarchical pathologist annotations across 4 branches and 8 follow-up question types; analyzed ∼8M cell images to isolate 2,653 disease-relevant cases
- Deployed Databricks training infrastructure with GPU auto-detection across 4 hardware tiers, image caching eliminating S3 I/O, and MLflow/Unity Catalog tracking; encoded 4 cytologic grading systems (Camus, Paes, Kiupel, Patnaik) as chain-of-thought signals

BigCircle (UPSAAS Technologies LLP), Gandhinagar, India
Artificial Intelligence Engineer — January 2025 – August 2025
- Developed end-to-end Deep Research pipeline using Python orchestrating ML prompt generation, Firecrawl API for web scraping, ChatGPT API summarization, graph visualization with Matplotlib and Seaborn, and automated Typst PDF reports; optimized API call efficiency reducing processing time by 75%
- Engineered pagination and authentication systems using JavaScript and Next.js, accelerating page load times by 40% with Docker containerization for 500+ concurrent sessions
- Collaborated with 5-member Agile team using Git for version control; performed code reviews to improve code quality

KEY PROJECTS & PUBLICATIONS

Next-Step: Full-Stack School Management Platform
Academic Project — May 2026
- Built async REST API with FastAPI exposing 20+ endpoints across 10 router modules backed by 9 SQLAlchemy ORM models (cascade deletes, JSON columns, unique constraints) with Alembic migrations on PostgreSQL and Celery/Redis with Flower monitoring for background tasks
- Implemented JWT authentication using Argon2 password hashing and Google OAuth 2.0 with PKCE flow, supporting Admin, Teacher, and Student roles via factory-pattern role-checking middleware
- Crafted cross-platform mobile app (iOS, Android, Web) with React Native and Expo using TypeScript, Zustand state management, file-based routing, and Google Drive integration for document management

MorphoCLIP: Cell Microscopy and Text Contrastive Learning
Research Capstone — April 2026
- Led 3-person team building dual-encoder contrastive learning system aligning cell microscopy with natural-language perturbations, achieving 24.3% Recall@10 on 817-perturbation retrieval, 3.3× over CellCLIP using 8M parameters versus 1.48B (185× reduction)
- Architected DINOv3 vision backbone paired with BioClinical ModernBERT text encoder, custom CrossChannelFormer fusing 5 fluorescence channels, and Cross-Well Alignment for batch-effect correction; enabled joint training on drug, CRISPR, and ORF perturbations
- Compressed 3.31 TiB of raw microscopy into 153 GB of cached DINOv3 embeddings, cutting per-epoch training to 3–5 min on single RTX 5080 across 3M+ images, 303 compounds, and 160 genes from CPJUMP1 benchmark

Hybrid ResNet-ViT for Skin Cancer Classification
Published in IEEE — July 2025
- Hybrid TensorFlow architecture combining frozen ResNet50 with four-head Vision Transformer blocks and Global Average Pooling for seven-class skin lesion classification, achieving 96.3% accuracy and macro F1 of 0.961 on HAM10000 (10,015 images, Albumentations preprocessing); presented at IEEE World Conference to 100+ attendees